MONTHLY RESEARCH REPORT

Month	Month 2 / From 01/06/2026 - 01/07/2026
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Research Theme	**Local LLM-Assisted Railway Terminology Extraction**
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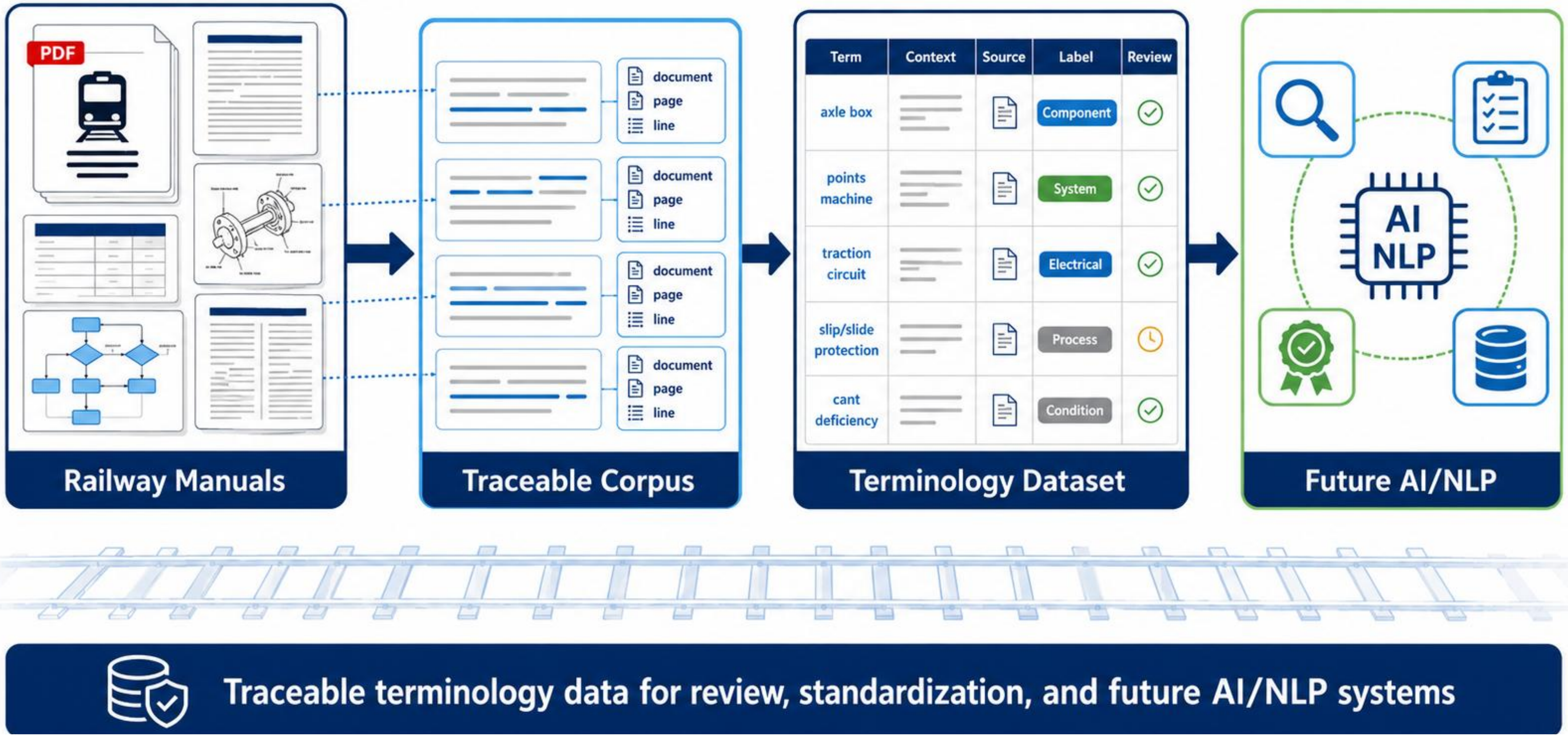
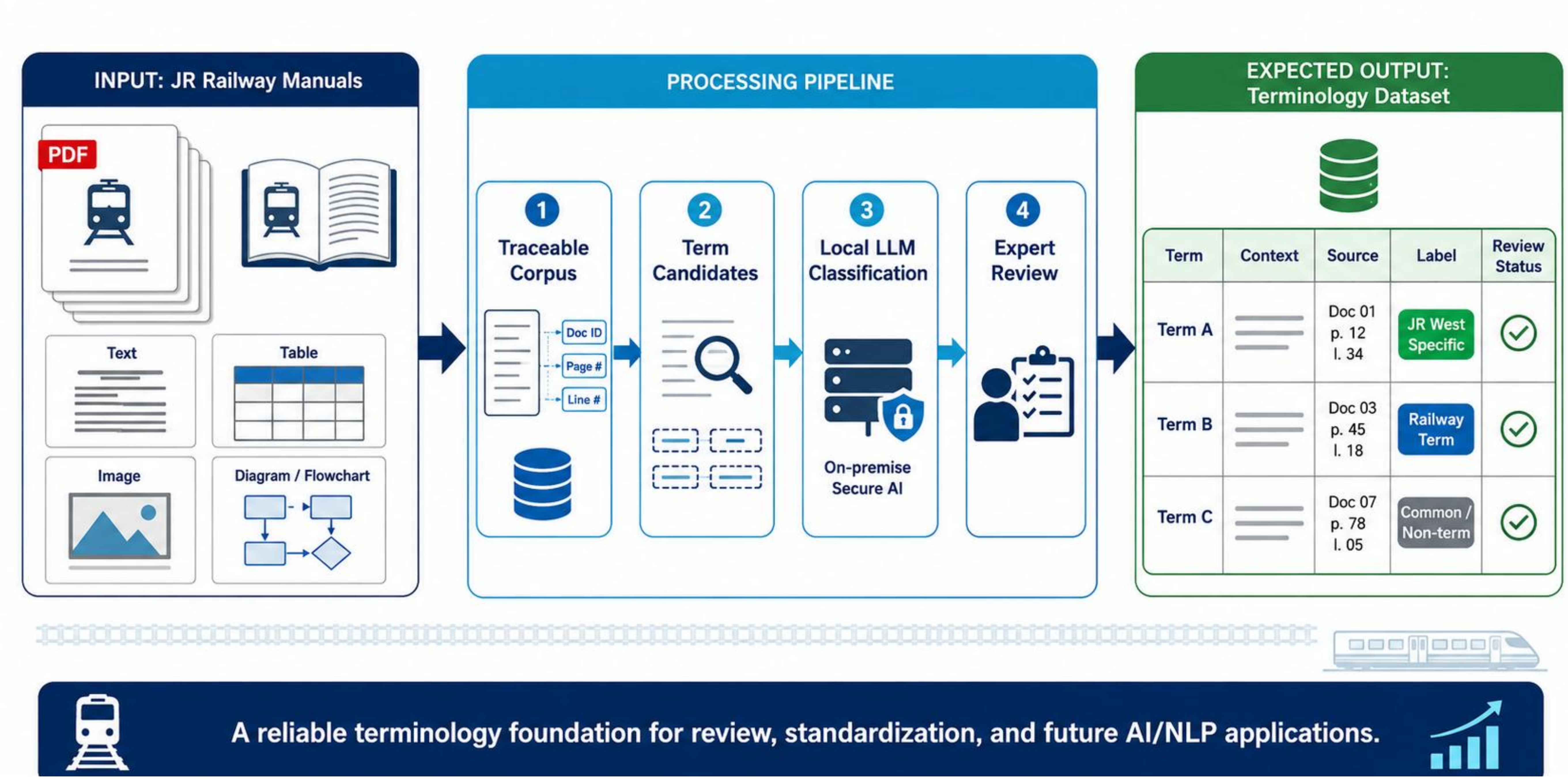
ONE LOVE. ONE FUTURE.

Project Vision: From Baseline Inference to Review-Ready Data



Build a traceable Japanese railway terminology workflow
where human-reviewed inference becomes future training data.

Month 2



This month validates the complete local inference loop before any fine-tuning: documents -> candidates -> Qwen JSON -> CSV review.

System Architecture

Inference first, human review second, supervised fine-tuning only after labels are reliable.

Phase 1
Document Ingestion
Input: Japanese railway PDFs
Output: Page-aware text
- PyMuPDF extraction
- NFKC normalization
- Page traceability

Phase 2
Candidate Selection
Input: Page-aware chunks
Output: Ranked chunks
- Domain dictionary matches
- Common-word negatives
- Generic-term filtering

Phase 3
Local Qwen Inference
Input: Chunk + dictionary signals
Output: Structured JSON
- known_term / new_candidate
- Exact context copy
- JSON repair and validation

Phase 4
Human Review -> SFT
Input: Draft terminology CSV
Output: Reviewed JSONL
- Accept / reject / correct
- Reviewer notes
- QLoRA only after review

Traceability: document -> page -> chunk -> term -> review decision.

Focus of Month 2

The priority is to validate inference quality and create a practical review loop, not to fine-tune prematurely. Reviewed outputs will become the supervised dataset for later QLoRA/SFT experiments.

Local Inference Baseline

Run Qwen locally on selected railway chunks.

- Qwen2.5-7B-Instruct-AWQ
- Deterministic JSON generation
- Checkpointed CSV output

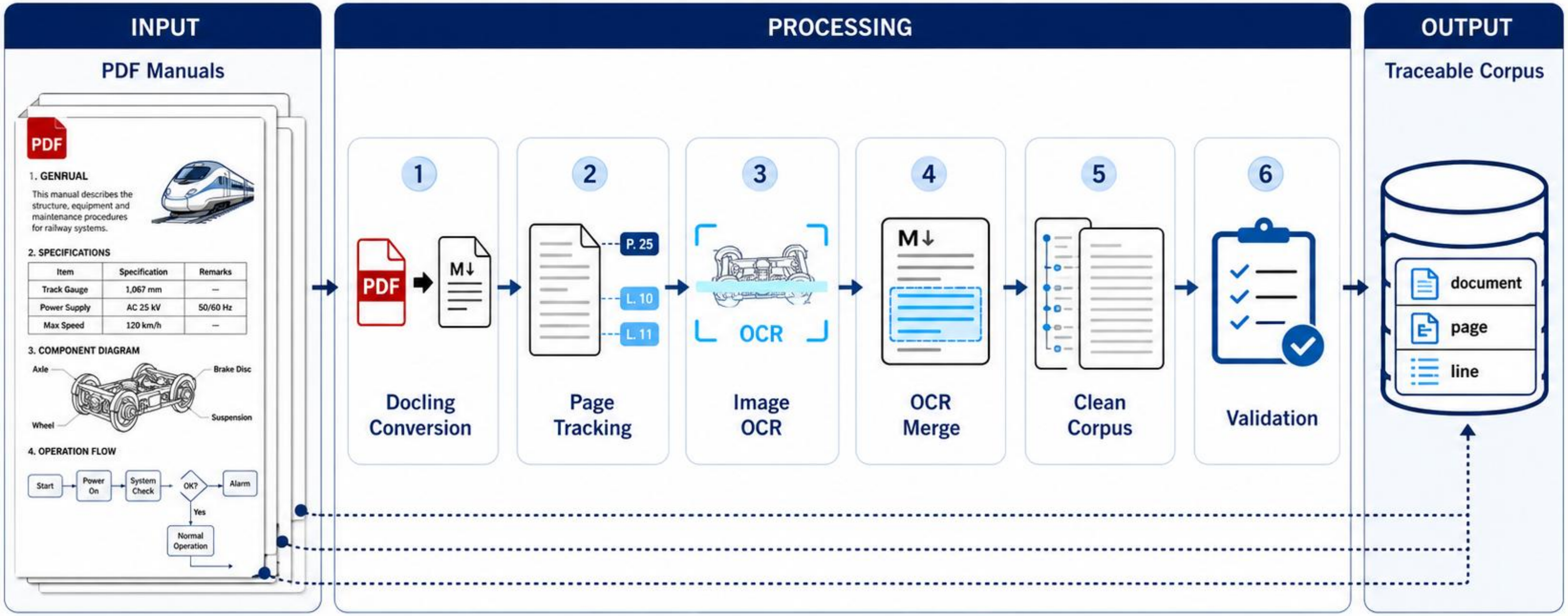
Reviewer-Facing Web App

Make terminology evidence easy to inspect.

- Multi-file upload and removal
- Page-aware PDF verification
- English / Japanese / Vietnamese UI

Phase 1: PDF to Traceable Chunks

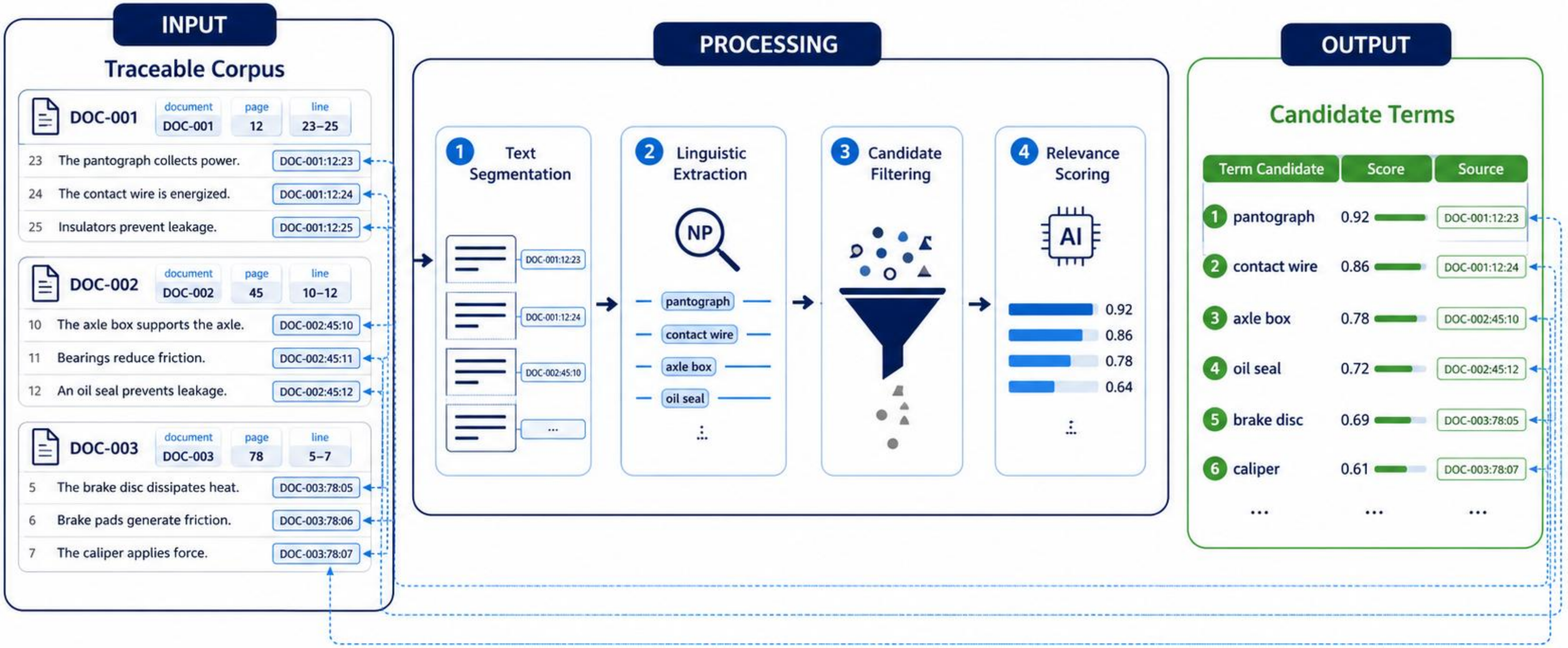
Normalize railway manuals while preserving document and page metadata.



Observed baseline: 4 PDFs -> 594 pages -> 683 chunks (maximum 700 characters, 80-character overlap).

Phase 2: Hybrid Candidate Extraction

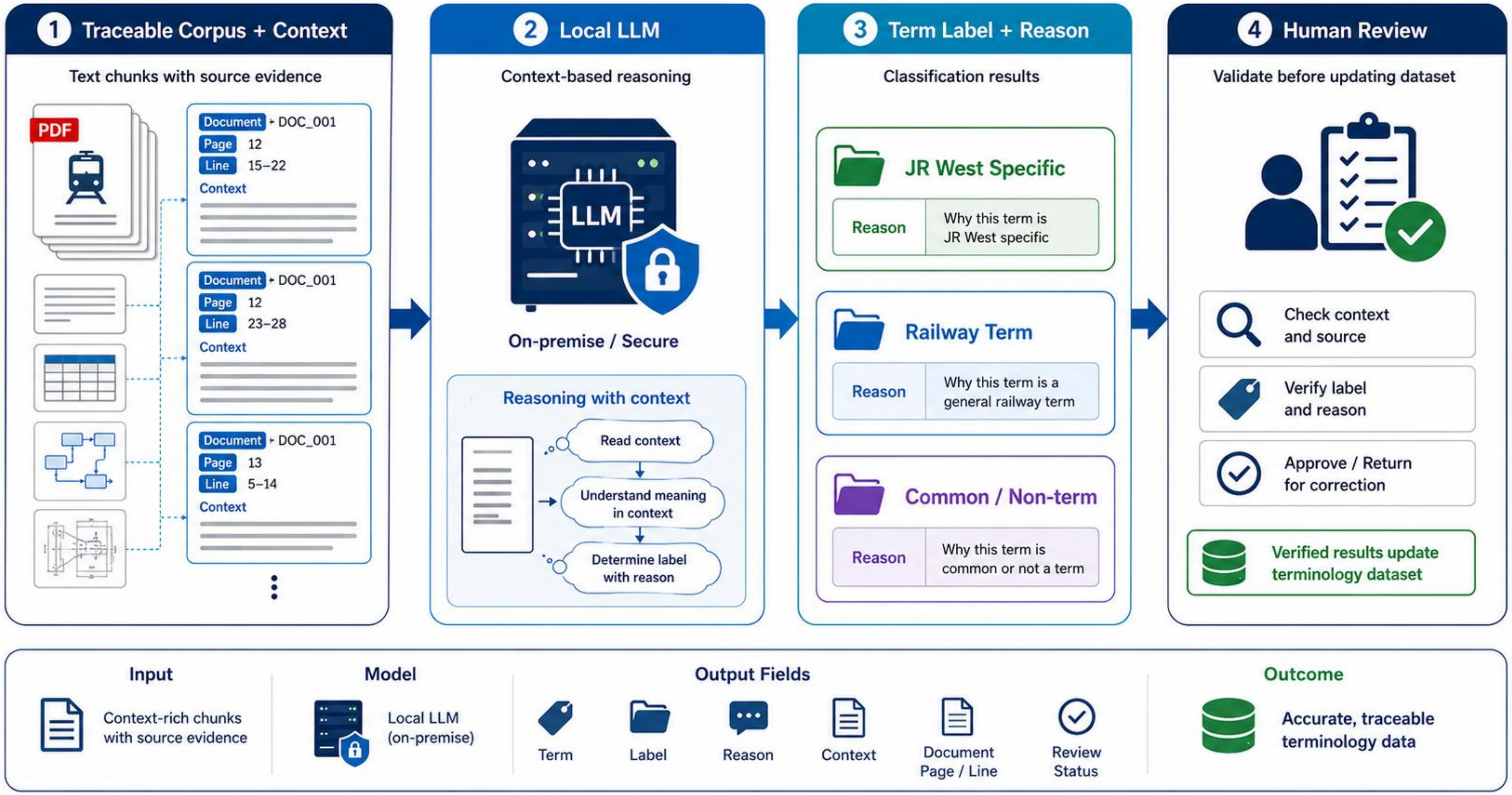
Dictionary evidence narrows the search space before expensive LLM inference.



8,991 usable domain terms + 13,353 common-word negatives -> 403 candidate chunks selected from 683.

Current Phase: Local Qwen Inference

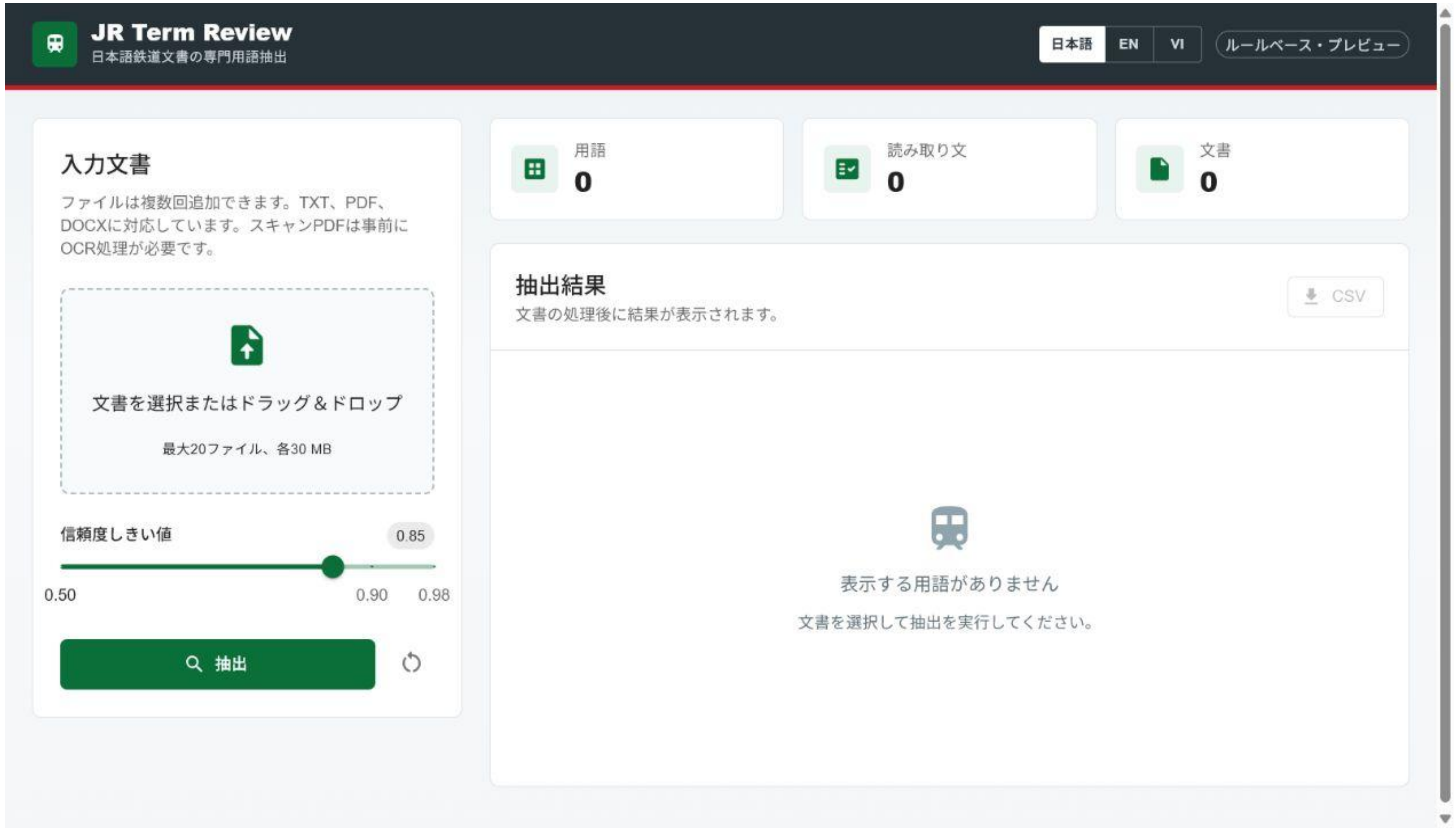
Use context and dictionary signals to produce reviewable term records, not final



Notebook checkpoint: 3 chunks completed and 36 raw rows saved; the current run is configured for 10 chunks.

Web Review Interface

The interface turns extraction output into a review workflow for railway-domain users



日本語 / EN / VI

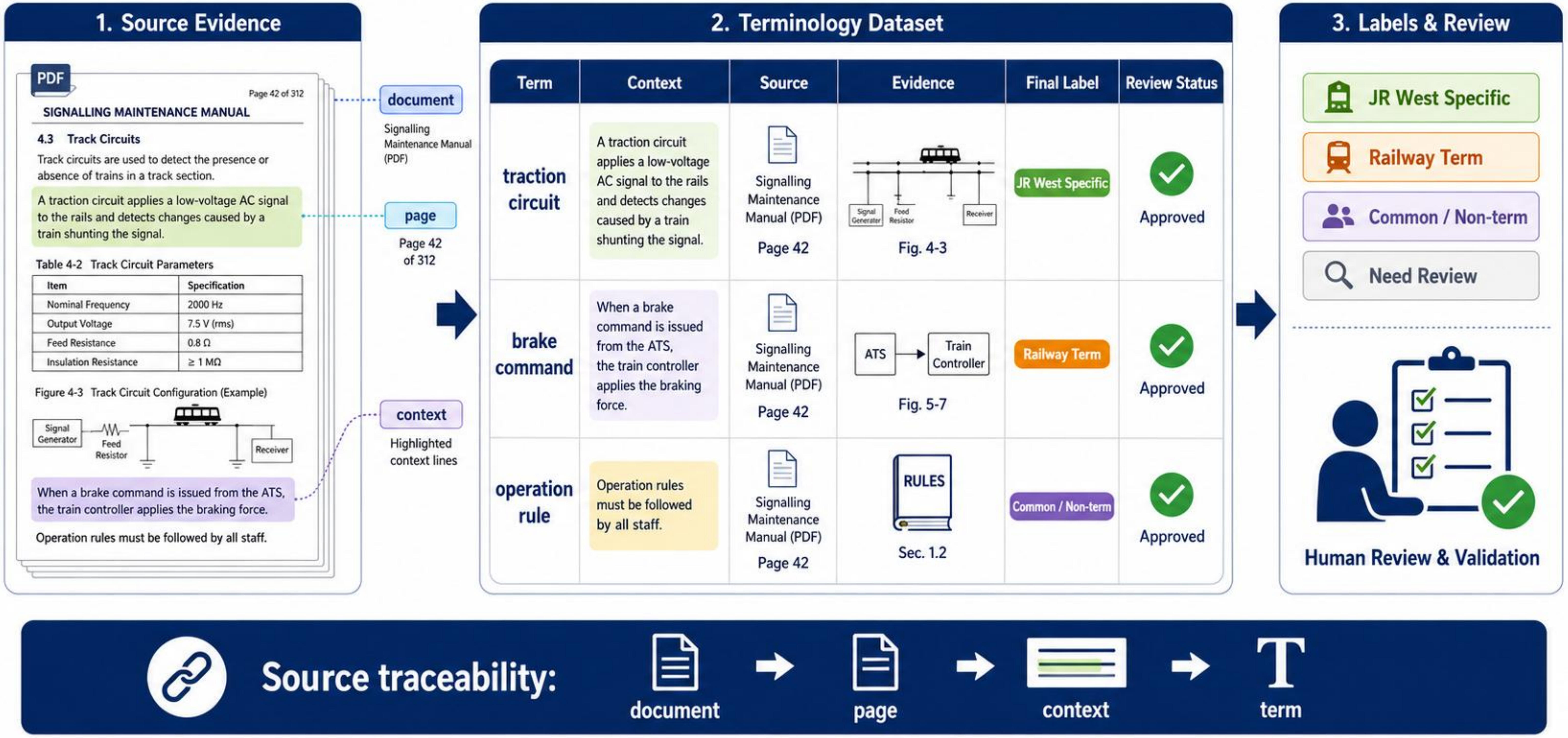
Multi-file input

CSV export + PDF evidence

Term, confidence, frequency, pages, source document and context

Human Review Before Fine-tuning

Inference creates draft labels; only reviewed labels are allowed into the SFT



Evaluation Roadmap

Progress from baseline inference to a measurable before/after fine-tuning comparison.

01 · CURRENT
Inference baseline
Run Qwen locally and export draft CSV.

02
Human review
Accept / reject / correct and add reviewer notes.

03
SFT dataset
Group reviewed labels by chunk into JSONL.

04
QLoRA / SFT
Train Qwen2.5-7B-Instruct, not the AWQ model.

05
Before / after
Compare precision, JSON stability, new candidates and review effort.

Training starts only after reviewed labels are sufficiently reliable (50-100 samples for a proof of concept).

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